



# **Explainable Deep Learning Framework for Pneumonia Detection and Classification Using Transfer Learning and Grad-CAM**

## **INTERIM REPORT 02**

G.C.M. Sewwandi

29002

Department of Computing

NSBM Green University

Bachelor of Science (Honours) in Computer Science

# Chapter 01 - Introduction

## 1.1 Chapter Overview

This chapter introduces the research project and establishes the context in which it is carried out. It opens with the clinical background of pneumonia diagnosis through chest X-ray imaging and the practical difficulties that motivate an automated and explainable approach. The problem is then framed at both a general and a discipline-specific level so that the precise research gap can be stated. From this gap the chapter derives a focused research question, explains the motivation behind the study, and sets out the aim and the objectives that direct the remaining work. A high-level picture of the proposed solution is presented to convey the intended workflow, followed by the hardware and software resources required, the boundaries of the project through an in-scope and out-of-scope definition, and a short summary that connects the chapter to the literature review that follows.

## 1.2 Problem Background

Pneumonia is one of the leading causes of illness and death worldwide and is responsible for millions of deaths each year, with children under five and elderly populations carrying a disproportionate share of that burden [1]. The disease is caused by a range of pathogens, including bacteria, viruses, fungi, and atypical organisms, and each of these presents with distinct radiographic and clinical characteristics that call for different treatment pathways. Distinguishing bacterial pneumonia, viral pneumonia, and COVID-19 from a normal lung is therefore not an academic exercise; it directly shapes antibiotic prescribing, isolation decisions, and the urgency of follow-up care. Chest X-ray imaging remains the most widely adopted first-line modality for pulmonary disease because it is accessible, inexpensive, quick to acquire, and available even in low-resource settings. Reliable interpretation, however, is far from straightforward. The visual signatures of COVID-19, bacterial pneumonia, viral pneumonia, and a healthy lung frequently overlap, and the findings can be subtle or non-specific in early infection. Radiologist availability is uneven across institutions and regions, and the high patient volumes typical of emergency and intensive-care settings can introduce diagnostic delays that worsen outcomes. Against this backdrop, automated and explainable tools offer meaningful support by reducing radiologist workload, improving the consistency of preliminary assessments, and providing a rapid second opinion best positioned to augment, rather than replace, the clinician [1].

## 1.3 Problem Statement

### 1.3.1 General Problem

At a high level, the core problem is that timely and accurate differentiation of pneumonia subtypes from normal lung conditions is difficult to guarantee under real clinical pressures. When interpretation is delayed or incorrect, the consequences cascade: treatment is started late, antibiotics may be prescribed where they are not warranted, infectious cases may not be isolated promptly, and vulnerable patients face an elevated risk of deterioration. These pressures are amplified wherever radiological expertise is scarce or where patient throughput is high, which describes a large proportion of healthcare settings rather than a narrow exception. The general problem this research addresses is therefore the absence of a dependable, transparent, and readily deployable tool that can assist clinicians with rapid preliminary chest X-ray interpretation.

### 1.3.2 Specific Problem

Within the field of deep-learning-based chest X-ray analysis, several discipline-specific shortcomings narrow the general problem into a concrete gap. First, the majority of published studies treat the task as a binary problem normal versus abnormal, or COVID-19 versus non-COVID-19 rather than the clinically meaningful four-way distinction among Normal, COVID-19, Bacterial Pneumonia, and Viral Pneumonia [2], [3], [4]. Second, the large benchmark datasets that have driven progress, such as ChestX-ray8 [5] and CheXpert [6], are designed for multi-label thoracic abnormality recognition rather than focused pneumonia subtype classification. Third, although strong architectures exist, there are few controlled comparisons of DenseNet121, EfficientNetV2B3, and EfficientNetV2S trained on a single dataset with an identical pipeline, which makes it hard to attribute performance differences to the architectures themselves. Fourth, many models behave as black boxes; without interpretability, clinicians cannot confirm that a prediction is grounded in genuine pathology rather than in spurious image artefacts, and explainability, where it is offered at all, tends to be a detached visualisation rather than a feature woven into the clinical workflow. Finally, complete and deployable decision-support systems those that combine secure access,

role-based dashboards, historical prediction management, and automated reporting are largely absent from the published literature. The specific research gap this project addresses is the lack of an end-to-end, explainable, four-class chest X-ray classification system that is engineered for real clinical deployment.

## 1.4 Research Question

The study is guided by a single principal research question:

**How can an explainable deep learning framework be designed, implemented, and deployed to accurately classify chest X-ray images into Normal, COVID-19, Bacterial Pneumonia, and Viral Pneumonia while providing clinically meaningful interpretability and practical, secure decision-support functionality?**

This principal question is supported by three sub-questions:

- Which of the candidate transfer-learning architectures: DenseNet121, EfficientNetV2B3, or EfficientNetV2S offers the most suitable balance of accuracy, interpretability, and deployment efficiency for four-class chest X-ray classification?
- How can Grad-CAM explainability be integrated as a workflow-embedded feature rather than an isolated visualisation, so that each prediction is accompanied by trustworthy visual evidence?
- How can the trained model be delivered as a secure, mobile-friendly platform that supports prediction history, automated multilingual reporting, and a safeguard against untrustworthy inputs?

## 1.5 Research Motivation

The motivation for this work arises from a persistent disconnect between what deep learning can achieve in a research setting and what is actually usable at the point of care. A model that reports a high accuracy figure in a notebook does little for a clinician who needs a result that is explainable, auditable, and embedded in a usable interface. Two observations shaped the project. The first is that clinical adoption depends on trust, and trust depends on transparency: a prediction is only useful when the clinician can see why the model reached it and can check that reasoning against the visible anatomy. The second is that the supporting machinery around a model secure login, prediction history, structured reporting, and access on ordinary mobile devices is what turns a classifier into a tool a clinic could realistically use. Building a system that takes interpretability and deployment as seriously as predictive accuracy was the central driver of this research.

## 1.6 Research Aim

The aim of this research is to design, develop, and evaluate an explainable, end-to-end clinical decision-support system that classifies chest X-ray images into four diagnostic categories using transfer-learned convolutional neural networks, integrates Grad-CAM visual explanations directly into the diagnostic workflow, and delivers this capability through a secure, mobile-friendly web platform with prediction history, automated multilingual report generation, and safeguards that withhold predictions on inputs the model was never trained to interpret.

## 1.7 Research Objectives

To achieve this aim, the following research-specific objectives are defined:

- 1.7.1 To identify the clinical and technical requirements of explainable four-class chest X-ray classification by reviewing the relevant literature, surveying the publicly available chest X-ray datasets, and establishing the gaps that an end-to-end decision-support system must close.
- 1.7.2 To analyse three modern convolutional architectures DenseNet121, EfficientNetV2B3, and EfficientNetV2S together with appropriate preprocessing, augmentation, and class-imbalance strategies, and to analyse the explainability and deployment requirements that a clinical tool must satisfy.
- 1.7.3 To design and develop a complete system comprising a four-class transfer-learning inference pipeline with integrated Grad-CAM, a Next.js web frontend, a PHP REST backend secured with JSON Web Tokens, a Python inference microservice, a relational database, an out-of-distribution safeguard, automated trilingual (English, Sinhala, and Tamil) PDF reporting, and a platform-assistance chatbot.
- 1.7.4 To evaluate the trained models using accuracy, precision, recall, F1-score, and confusion-matrix analysis, to assess the clinical relevance of the Grad-CAM explanations, and to verify that the deployed system delivers its intended decision-support functionality.

## 1.8 Rich Picture of the Proposed Solution

The proposed solution, named LungVision AI, is a web-based lung disease detection platform with mobile-friendly, progressive web application style delivery. Figure 1 illustrates the intended end-to-end workflow. A registered user uploads a chest X-ray through the web interface. The backend forwards the image to a dedicated inference microservice, which preprocesses the image to the resolution and normalisation expected by the selected model, runs the convolutional network to obtain class probabilities, and produces a Grad-CAM heatmap that highlights the regions most responsible for the prediction. Before the result is trusted, a post-hoc out-of-distribution check decides whether the upload actually resembles the chest X-rays the model was trained on; if it does not, the diagnosis is withheld and the user is warned. The class scores, predicted condition, confidence, Grad-CAM overlay, and out-of-distribution verdict are returned to the backend, persisted in the database, and presented on the dashboard with confidence visualisation. Every prediction is stored as part of the patient's history, can be exported as a structured PDF report in English, Sinhala, or Tamil, and is accompanied by an optional chatbot that helps users navigate the platform. Access to all of this is governed by secure authentication and role-based permissions.

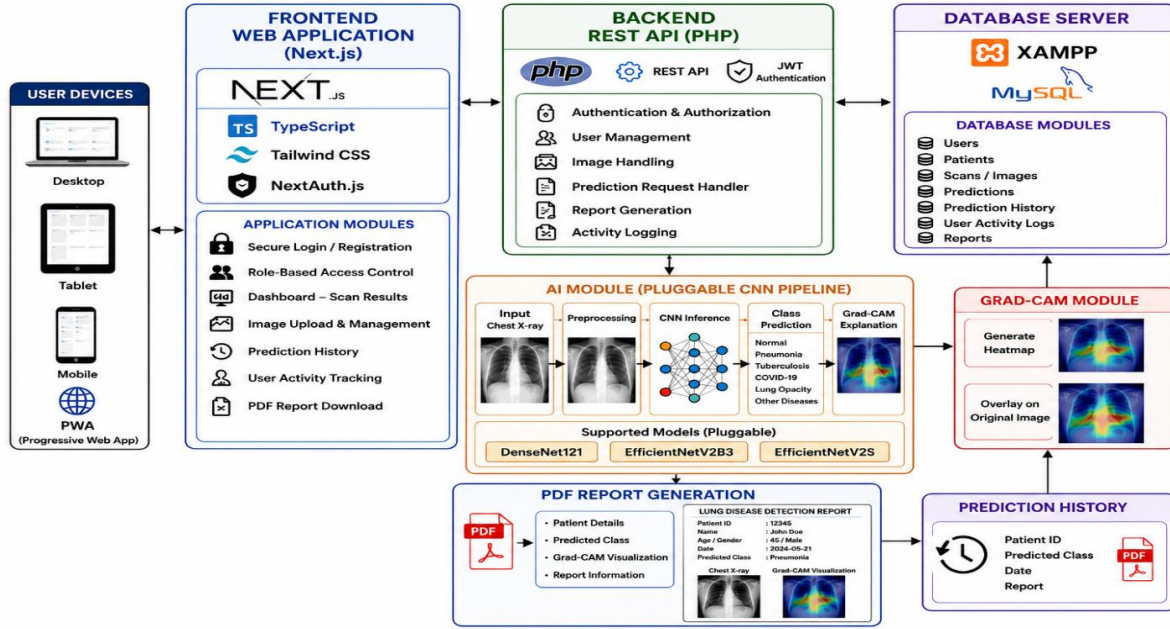


Figure 1. End-to-end workflow of the proposed LungVision AI decision-support platform.

## 1.9 Resource Requirements

### 1.9.1 Hardware

Model training requires a workstation equipped with a CUDA-capable GPU together with sufficient system memory and storage to hold the chest X-ray dataset and the resulting model checkpoints. For deployment and day-to-day inference the demands are far lighter: a standard server or development machine capable of running the Apache web server, the MySQL database, and the Python inference microservice is sufficient, and predictions can be served on the CPU. End users require only an ordinary desktop, tablet, or mobile device with a modern web browser, since all heavy computation takes place on the server side.

### 1.9.2 Software

The implementation draws on an open-source software stack. Model development uses Python with TensorFlow and Keras, served through a FastAPI inference microservice. The application backend is built on PHP 8 running under XAMPP with a MySQL database, exposing a REST API secured by JSON Web Tokens. The frontend is a Next.js 14 application written in TypeScript and styled with Tailwind CSS, using NextAuth.js for session handling and Recharts for confidence visualisation. The platform-assistance chatbot is powered by a hosted large language model accessed through a server-side key. Supporting tools include the Pillow imaging library for preprocessing, version control with Git, and phpMyAdmin for database administration.

## 1.10 Project Scope

The boundaries of the project are summarised in Table 1. The work concentrates on building and evaluating the four-class classification capability and the surrounding decision-support platform; it deliberately excludes activities, such as prospective clinical trials, that lie beyond the remit of an undergraduate research project.

**TABLE 1. PROJECT SCOPE DEFINING THE IN-SCOPE AND OUT-OF-SCOPE BOUNDARIES OF THE RESEARCH**

| In Scope                                                                                                                   | Out of Scope                                                                                                                  |
|----------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------|
| Four-class classification of chest X-rays into Normal, COVID-19, Bacterial Pneumonia, and Viral Pneumonia.                 | Diagnosis of pulmonary conditions outside these four categories, or of diseases from other imaging modalities (e.g. CT, MRI). |
| Comparative evaluation of DenseNet121, EfficientNetV2B3, and EfficientNetV2S under a unified training pipeline.            | Training of entirely novel network architectures from scratch without transfer learning.                                      |
| Grad-CAM explainability and a post-hoc out-of-distribution safeguard integrated into the workflow.                         | Pixel-level lesion segmentation or automated radiology report writing.                                                        |
| A deployable web platform with secure authentication, role-based access, prediction history, and trilingual PDF reporting. | Integration with hospital information systems, PACS, or electronic health records.                                            |
| Offline evaluation on a curated public chest X-ray dataset.                                                                | Prospective clinical trials and regulatory certification for autonomous diagnosis.                                            |

## 1.11 Chapter Summary

This chapter established the clinical and technical context for the research. It described the burden of pneumonia and the difficulty of reliable chest X-ray interpretation, framed the general and specific dimensions of the problem, and identified the gap that motivates an end-to-end, explainable, four-class decision-support system. From this gap it derived the research question, the motivation, the aim, and four research-specific objectives, and it outlined the proposed LungVision AI solution along with its resource requirements and scope. The next chapter reviews the literature that underpins these design choices and substantiates the identified research gap.

# Chapter 02 - Literature Review

## 2.1 Chapter Overview

This chapter reviews the body of work on which the proposed system rests and positions the research against it. Rather than cataloguing studies in isolation, the review weighs each contribution against the requirements of a deployable, explainable, four-class chest X-ray classifier and records where the present project agrees with, departs from, or extends the existing literature. The chapter begins with a conceptual map that explains how the review is organised. It then surveys the clinical and computational domain at a high level, compares existing systems and frameworks, and undertakes a deeper technological analysis of the algorithms, designs, and workflows that are relevant to the project. A reflective section closes the chapter by drawing the threads together and justifying the research gap.

## 2.2 Conceptual Map of the Literature

The reviewed literature is organised into three concentric layers that move from breadth to depth. The outermost layer, the domain overview, situates the work within medical imaging and deep learning and accounts for roughly a tenth of the discussion. The middle layer examines existing systems, frameworks, and designs for pneumonia and COVID-19 detection, comparing them critically and accounting for around a third of the discussion. The innermost and largest layer is a technological analysis that decomposes the field into its algorithmic, design, and workflow dimensions and accounts for the remaining majority of the review. Threaded through all three layers are two recurring concerns predictive performance and clinical trustworthiness and it is the tension between them that the reflection ultimately resolves into a statement of the research gap.

## 2.3 Domain Overview

Deep learning has become the dominant paradigm in medical image analysis. Convolutional neural networks learn a hierarchy of visual representations directly from data, progressing from low-level edges and textures to high-level structural patterns without the handcrafted feature engineering that earlier methods relied upon [7]. The foundational demonstration that deep networks can achieve strong image-recognition performance, given suitable architectures and training strategies, motivated a wave of clinical applications aimed at supporting diagnosis while easing the burden on specialists [1], [8]. Comprehensive surveys confirm that convolutional networks have been applied successfully to classification, detection, segmentation, and prognosis across medical imaging, while cautioning that careful methodology is needed to manage dataset bias, class imbalance, and spurious correlations [9]. Progress in chest X-ray analysis specifically has been accelerated by large public benchmarks. The ChestX-ray8 dataset established a hospital-scale, multi-label thoracic abnormality benchmark of more than one hundred thousand images across fourteen disease labels [5], and CheXpert extended this paradigm with uncertainty labels and expert comparison protocols [6]. During the pandemic, dedicated COVID-19 chest X-ray collections enabled the rapid development of automated screening models [10]. These resources are invaluable, yet their orientation towards multi-label thoracic findings means they do not directly serve the focused, mutually exclusive four-class distinction Normal, COVID-19, Bacterial Pneumonia, and Viral Pneumonia that is clinically meaningful for triage and treatment selection. This observation frames the rest of the review: the building blocks exist, but they have rarely been assembled into the specific configuration this project requires. A further characteristic of the domain, evident across these datasets, is the heterogeneity and imbalance of the available data. Chest X-rays are acquired on different equipment, under different protocols, and from different patient populations, so they vary considerably in resolution, contrast, and intensity, and the disease classes are seldom evenly represented because they follow natural prevalence rather than a balanced experimental design. Both properties have practical consequences for any classifier trained on such data: without careful normalisation a model can latch onto acquisition artefacts rather than pathology, and without explicit handling of imbalance it can drift toward the majority classes. The literature on deep learning in medical imaging repeatedly stresses that methodological care around these issues is as decisive as the choice of network [9], a lesson that directly shapes the preprocessing and class-weighting decisions described in the methodology.

## 2.4 Existing Systems, Frameworks, and Designs

The most influential early system in this space is CheXNet, which used a densely connected network to achieve radiologist-level pneumonia detection on chest X-rays and demonstrated convincingly that deep models could match expert performance on a focused task [11]. CheXNet set the template for much of the work that followed, but it framed pneumonia detection as a binary problem and did not separate the pneumonia subtypes that carry different clinical implications. In my assessment, this is the recurring limitation across the field: strong results are reported on coarse label sets that do not map cleanly onto the decisions a clinician actually has to make. The COVID-19 pandemic produced a large volume of detection systems built on transfer learning. Apostolopoulos and Mpesiana applied pretrained convolutional networks to automatically detect COVID-19 from X-ray images and reported encouraging accuracy with relatively small datasets [2]. Ozturk et al. proposed an end-to-end deep model for automated COVID-19 detection that also offered a degree of localisation [3], and Narin et al. compared several pretrained convolutional networks for coronavirus detection from chest X-rays [4]. Collectively these studies confirmed that transfer learning is an effective response to the scarcity of annotated medical data, but they share three weaknesses that are directly relevant to the present work. They are predominantly binary or three-class formulations that set COVID-19 against a small number of alternatives; they were trained and evaluated on heterogeneous datasets with differing preprocessing and evaluation protocols, which makes cross-study comparison unreliable; and they generally stop at the model, offering little in the way of a deployable, secured, and auditable system around it. It is worth dwelling on why the binary framing that dominates these systems is a substantive limitation rather than a mere labelling choice. A model that separates abnormal from normal, or COVID-19 from everything else, leaves untouched the very distinction that most influences management: bacterial pneumonia typically warrants antibiotics, viral pneumonia and COVID-19 generally do not, and COVID-19 in particular carries isolation implications the others do not share. Collapsing these conditions into a single abnormal category therefore discards clinically actionable information. A four-class formulation restores that information, but it also makes the task harder, because the radiographic appearances of the three disease classes overlap far more with one another than any of them does with a healthy lung. This added difficulty is one reason the four-class problem remains under-explored, and it raises the premium both on architectures capable of capturing subtle, multi-scale differences and on evaluation that looks beyond headline accuracy to per-class behaviour.

Beyond single-model systems, ensemble strategies have been explored to push accuracy higher. Huang and Liao combined a stacking ensemble with an attention-enhanced EfficientNetV2 to classify multiple chest diseases, including COVID-19, and reported competitive performance on a multi-class problem [12]. This line of work is instructive because it demonstrates the viability of the EfficientNetV2 family for multi-class chest disease classification and so supports the decision to include EfficientNetV2 variants in the present comparison. At the same time, ensembles increase computational cost and complexity, which works against the deployment-oriented goals of this project; a single well-chosen architecture that can be served efficiently on modest hardware is a better fit for a tool intended to run alongside an ordinary web stack. Taken together, the existing systems establish feasibility but leave a clear opening. None of the reviewed designs simultaneously delivers a clinically meaningful four-class distinction, a controlled comparison of competitive architectures under one pipeline, explainability that is embedded in the workflow rather than presented as an afterthought, and the deployment scaffolding authentication, role-based dashboards, prediction history, and automated reporting that a usable clinical tool demands. The proposed system is designed precisely around closing that combined gap rather than improving any single metric in isolation.

## 2.5 Technological Analysis

This section analyses the technologies underpinning the project across three dimensions: the algorithms and network architectures, the system and software design, and the end-to-end workflow. The aim is not only to describe these technologies but to justify, against the alternatives, why each was selected for LungVision AI.

### 2.5.1 Algorithmic Analysis

Modern medical image classification rests on a sequence of landmark architectural advances in computer vision. The introduction of residual learning resolved the degradation problem that had previously made very deep networks difficult to train; skip connections allowed networks of more than a hundred layers to be optimised stably and delivered substantial gains over earlier designs [13]. Earlier milestones deep convolutional networks with non-linear activations and dropout regularisation, multi-scale inception modules, and very deep networks built from small convolutional filters together established the representational capacity and the training practices that modern medical classifiers inherit. Visualisation studies that revealed how intermediate convolutional layers respond selectively to image patterns provided early insight into the learned feature hierarchy and foreshadowed the explainability techniques used today [14]. DenseNet builds on this lineage with dense connectivity, in which each layer receives the feature maps of all preceding layers within a dense block. This promotes feature reuse, improves gradient flow, and reduces parameter redundancy relative to conventional architectures [15]. These properties are well matched to chest X-ray analysis, where discriminating between opacity distributions, peribronchial thickening, airspace consolidation, and ground-glass opacities depends on combining information across multiple spatial scales. The effectiveness of densely connected networks in exactly this setting was demonstrated by CheXNet, which reached radiologist-level pneumonia detection [11], and it is this combination of fine-grained pattern capture and parameter efficiency that makes DenseNet121 a strong candidate for the present task.

The EfficientNet family approaches scaling from a different angle. Rather than enlarging depth, width, or resolution in isolation, it applies a principled compound-scaling rule that grows all three dimensions together using a fixed set of coefficients [16]. EfficientNetV2 refined this family by introducing Fused-MBConv blocks in the earlier stages and progressive learning with adaptive regularisation, which improved training efficiency without sacrificing accuracy [17]. EfficientNetV2B3 offers a particularly attractive balance of accuracy and computational cost for deployment-oriented applications, while the larger EfficientNetV2S allows an assessment of whether additional capacity and a greater proportion of Fused-MBConv blocks translate into meaningful gains. Including both a cross-family contrast (DenseNet versus EfficientNetV2) and a within-family contrast (EfficientNetV2B3 versus EfficientNetV2S) is what makes the comparison informative rather than arbitrary. Because annotated medical images are costly and time-consuming to obtain, transfer learning is central to the algorithmic strategy. Pan and Yang formally framed transfer learning as the reuse of knowledge from a source domain to improve learning in a related target domain [18], and Tajbakhsh et al. provided the empirical counterpart, showing that fine-tuning pretrained networks is consistently more effective than training from scratch across a range of medical imaging tasks [19]. Although ImageNet [20] contains natural rather than medical images, its early convolutional layers learn generic edges, textures, and gradients that transfer well, leaving only the higher-level representations to be specialised through fine-tuning. This evidence directly informs the two-stage transfer-learning protocol adopted in the methodology.

Explainability completes the algorithmic picture. Class activation mapping introduced the idea of localising the discriminative regions that drive a network's decision [14], and Grad-CAM generalised this by using the gradients

flowing into the final convolutional layer to produce class-discriminative heatmaps without any architectural modification [21]. For a clinical tool this is essential rather than cosmetic: surveys of explainable artificial intelligence in medicine stress that interpretability is a precondition for clinical acceptance and responsible deployment [22], and broader treatments of the subject argue that transparency is indispensable wherever decisions carry high stakes [23]. The consistent message that artificial intelligence should augment the clinician with interpretable, workflow-integrated outputs rather than replace expert judgement [1], [8] is the principle around which the explainability design of this project is built. The progression from class activation mapping to Grad-CAM also clarifies a design choice that matters in practice. The original class activation mapping method localised discriminative regions but required a particular network structure, with a global average pooling layer feeding directly into the classifier, which constrained the architectures it could be applied to [14]. Grad-CAM removed that constraint by deriving its importance weights from gradients rather than from a fixed layer arrangement, so it can be attached to a pretrained DenseNet121 or EfficientNetV2 backbone without redesigning the network [21]. For a project that deliberately compares heterogeneous architectures this generality is precisely what is needed, because the same explanation mechanism can be applied uniformly across all three models and the explainability comparison can be kept as fair as the accuracy comparison. It is this architectural neutrality, as much as the visual quality of the heatmaps, that makes Grad-CAM the appropriate choice here.

### 2.5.2 Design Analysis

The design analysis turns from algorithms to the architecture of the system that surrounds them. A recurring shortcoming of the reviewed literature is that the model is treated as the end product; here it is treated as one component within a layered, service-oriented design. The chosen architecture separates concerns into three cooperating tiers: a presentation tier, an application and data tier, and a dedicated inference tier. The presentation tier is a Next.js application written in TypeScript and styled with Tailwind CSS. A component-based framework with server-side rendering was preferred over a traditional multi-page application because it supports a responsive, app-like experience across desktop, tablet, and mobile devices while keeping the codebase maintainable. Session handling is delegated to NextAuth.js, and confidence scores are visualised with a charting library, giving clinicians an immediate sense of how decisive a prediction is. The application tier is a PHP REST backend running under XAMPP and backed by a MySQL relational database; it is responsible for authentication with JSON Web Tokens, role-based authorisation, persistence of scans and activity logs, and orchestration of the report and chatbot endpoints. Keeping this tier deliberately conventional makes the system easy to host on the ubiquitous LAMP-style stack found in many institutions.

The most important design decision is the separation of inference into its own Python microservice. Deep-learning inference depends on TensorFlow and Keras, which sit far more naturally in the Python ecosystem than in PHP. By exposing the trained models behind a small FastAPI service, the design keeps the heavyweight machine-learning dependencies isolated from the web application, allows the model to be scaled or replaced independently, and supports a pluggable pipeline in which DenseNet121, EfficientNetV2B3, or EfficientNetV2S can be selected according to deployment needs. This decoupling is precisely the kind of deployment-aware design that the surveyed single-model studies omit. The relational schema reflects the same philosophy: users carry a role that governs access, every scan records its predicted class, per-class confidence scores, the model used, the Grad-CAM image, and an out-of-distribution verdict, and an activity log captures actions for accountability.

### 2.5.3 Workflow Analysis

The workflow analysis examines how these components cooperate to turn an uploaded image into a trustworthy, reportable result. The pipeline begins with secure authentication and a role-appropriate dashboard. When an image is uploaded, the backend forwards it to the inference service, which applies the model-specific preprocessing, runs the network to obtain the four class probabilities, and generates the Grad-CAM overlay in a single pass. A critical addition, absent from the reviewed systems, is an out-of-distribution gate placed between prediction and presentation. Because a softmax classifier is closed-world and will always assign one of its four labels even to an image that is not a chest X-ray at all a post-hoc check is used to decide whether the input genuinely resembles the training distribution. When it does not, the diagnosis is withheld and the user is told why, which is the kind of safety behaviour a clinical tool ought to exhibit. Once a result is accepted, the workflow persists it so that it becomes part of the patient's prediction history, available for later review and trend visualisation. From any stored result the user can generate a structured PDF report containing the patient details, the predicted condition, the confidence scores, and the Grad-CAM visualisation, and can do so in English, Sinhala, or Tamil a feature that directly addresses accessibility for the local clinical context and that is unusual among published systems. A platform-assistance chatbot runs alongside the workflow to help users understand how to upload images and interpret results, while

remaining explicitly outside the role of a medical advisor. The result is a workflow in which classification, explanation, safety checking, history management, and reporting are not separate experiments but stages of one coherent clinical process.

The reporting stage deserves particular emphasis because it is where the system reaches beyond the screen and into the clinical record, and because accessibility is rarely treated as a first-class concern in the literature. Generating the report in English, Sinhala, and Tamil acknowledges that a tool intended for the local healthcare context must communicate with clinicians and patients in the languages they actually use, and that a result which cannot be read and acted upon has limited value however accurate it may be. Producing the report on the server, with the predicted condition, the confidence scores, and the Grad-CAM overlay laid out in a consistent structure, also makes each prediction auditable after the fact rather than ephemeral. Coupled with the prediction history and the activity log, this turns the platform from a one-shot classifier into a system that keeps a coherent, reviewable account of the decisions it has supported the kind of accountability that responsible deployment of clinical artificial intelligence requires [8], [23].

## 2.6 Reflection

Reading the literature against the requirements of a deployable clinical tool brings a consistent pattern into focus. The field has convincingly established that deep convolutional networks, initialised through transfer learning, can detect pneumonia and COVID-19 from chest X-rays with impressive accuracy [11], [2], [3], [4], and it has produced both the architectures [13], [15], [16], [17] and the explainability techniques [21], [14] needed to do so responsibly. What it has largely not done is bring these pieces together. Five gaps recur across the recent and otherwise strong literature: most studies remain binary rather than addressing the clinically meaningful four-class distinction; that four-class distinction among Normal, COVID-19, Bacterial Pneumonia, and Viral Pneumonia is under-explored within a single unified framework; few studies compare DenseNet121, EfficientNetV2B3, and EfficientNetV2S under one dataset and pipeline; explainability is typically an add-on visualisation rather than a workflow-integrated feature linked to patient records; and deployment concerns such as secure authentication, role-based dashboards, prediction history, and automated reporting are frequently absent. These five gaps, drawn from literature that is for the most part recent, jointly define the research gap this project addresses. The contribution of LungVision AI is therefore not a single new algorithm but the deliberate integration of a controlled four-class architecture comparison, workflow-embedded Grad-CAM explainability, an out-of-distribution safeguard, and a secure, mobile-friendly deployment with multilingual reporting into one end-to-end decision-support system. The methodology chapter that follows sets out how this integrated system is designed, built, and evaluated.

# Chapter 03 - Methodology

This chapter sets out the research methodology and the concrete technical approach used to build and evaluate LungVision AI. It first establishes the philosophical and strategic framing of the research the paradigm, approach, strategy, and fact-collection mechanisms before mapping the execution of the work onto a structured design-science workflow. The detailed engineering methodology, covering the dataset, preprocessing, model training, explainability, and system deployment, is presented within that workflow, followed by the project-management approach, the timeline, the ethical considerations, and a closing summary.

## 3.1 Research Paradigm

The research is conducted within a pragmatic paradigm that combines a positivist stance toward measurement with a design-science orientation toward building an artefact. The positivist element is appropriate because the central claims of the work that one architecture classifies chest X-rays more accurately than another, for instance are objective and empirically testable through controlled experiments and quantitative metrics. The pragmatic element acknowledges that the ultimate purpose of the research is not knowledge for its own sake but a working tool that solves a real clinical problem, which shifts the emphasis toward what demonstrably works in practice. This framing justifies an approach in which a system is designed and constructed and its components are then measured against objective criteria.

## 3.2 Research Approach

The study follows a primarily deductive approach embedded within design-science research. The literature yields clear, testable expectations that transfer learning will outperform training from scratch under data constraints, and

that competitive architectures will differ in measurable ways on a unified pipeline and the experiments are designed to confirm or refute these expectations rather than to induce theory from open-ended exploration. Surrounding this deductive core is the constructive logic of design science: an artefact is specified, built, demonstrated, and evaluated, and the lessons from evaluation feed back into its refinement.

### 3.3 Research Strategy

Two complementary strategies are used. The first is controlled experimentation: the three candidate architectures are trained and tested under an identical dataset split, preprocessing regime, augmentation policy, optimisation configuration, and evaluation protocol, so that any observed difference in performance can be attributed to the architectures themselves rather than to confounding experimental conditions. The second is prototyping through design-science research: the trained model is wrapped in a complete, functioning software system whose components the inference microservice, the REST backend, the relational database, the web frontend, the explainability and safety layers, and the reporting and chatbot features are progressively built and demonstrated. Together the two strategies cover both the scientific question of which model is best and the engineering question of how to deliver it usefully.

### 3.4 Fact Collection Mechanisms

Because the research concerns image classification rather than human subjects, the primary facts are drawn from secondary data and from controlled experiments rather than from surveys or interviews. Three mechanisms are used. First, a structured review of the academic literature establishes the state of the art, the available architectures and explainability techniques, and the research gap. Second, publicly available, de-identified chest X-ray datasets supply the labelled images used to train and evaluate the models. Third, the experiments themselves generate quantitative evidence accuracy, precision, recall, F1-score, confusion matrices, and the qualitative Grad-CAM overlays which constitutes the principal empirical record against which the system is judged.

### 3.5 Research Methodology Execution Workflow

The execution of the research is organised as a design-science workflow whose stages map directly onto the activities of problem identification, relevance justification, comparative analysis, objective definition, design and development, and evaluation and communication. Table 2 summarises how each stage is addressed in this project, after which the design and development stage which carries the detailed engineering methodology is expanded in full.

**TABLE 2. MAPPING OF THE DESIGN-SCIENCE EXECUTION WORKFLOW STAGES TO HOW EACH IS ADDRESSED IN THIS RESEARCH**

| Workflow Stage                             | How it is Addressed in this Research                                                                                                                                                                                     |
|--------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Problem Identification                     | Establishing the difficulty of reliable, timely chest X-ray interpretation and the absence of an explainable, deployable four-class decision-support tool (Sections 1.2-1.3).                                            |
| Relevance Justification                    | Grounding the problem in the clinical burden of pneumonia and the need for interpretable, workflow-integrated AI support [1], [8], confirming the work addresses a genuine need.                                         |
| Comparative Analysis and Gap Justification | Critically comparing existing systems and architectures in the literature review to derive the five-part research gap (Sections 2.4-2.6).                                                                                |
| Definition and Finalisation of Objectives  | Translating the gap into the four research objectives of identifying, analysing, designing/developing, and evaluating the system (Section 1.7).                                                                          |
| Design, Development, and Data Management   | Preparing the dataset, preprocessing and augmenting it, training the three architectures with transfer learning, integrating Grad-CAM and out-of-distribution detection, and building the full platform (Section 3.5.5). |
| Evaluation and Communication               | Measuring the models with standard metrics, assessing explanation quality, validating system functionality, and communicating the outcomes through this report (Section 3.5.6).                                          |

#### 3.5.1 Problem Identification

The first stage fixed the problem precisely: classifying chest X-rays into four mutually exclusive categories: Normal, COVID-19, Bacterial Pneumonia, and Viral Pneumonia in a way that is accurate, interpretable, and deployable.

Identifying the closed-world behaviour of softmax classifiers as a safety risk was part of this stage and led to the inclusion of an out-of-distribution safeguard in the design.

### 3.5.2 Relevance Justification

The relevance of the problem is anchored in the clinical reality that pneumonia subtypes demand different treatment, isolation, and follow-up decisions, so that a tool capable of rapid, transparent preliminary discrimination has tangible value [1]. The justification also rests on the principle that decision support must be interpretable and embedded in the clinical workflow to be adopted at all [8], which the design takes as a binding constraint rather than an optional feature.

### 3.5.3 Comparative Analysis and Gap Justification

The comparative analysis carried out in the literature review distinguished the present work from prior systems and justified the research gap. By weighing CheXNet [11], the COVID-19 transfer-learning studies [2], [3], [4], and the EfficientNetV2 ensemble approach [12] against the requirements of a deployable four-class tool, this stage confirmed that no existing system combines a unified architecture comparison, embedded explainability, an input-safety safeguard, and a secured, reportable deployment.

### 3.5.4 Definition and Finalisation of Objectives

The identified gap was translated into the four research objectives stated in Section 1.7. Each objective is deliberately research-specific rather than generic: to identify the requirements and datasets, to analyse the candidate architectures and deployment needs, to design and develop the integrated system, and to evaluate it against quantitative and qualitative criteria. Finalising these objectives fixed the boundaries of the development and evaluation work.

### 3.5.5 Design, Development, and Data Management

This stage contains the core engineering methodology and is organised into the dataset, the preprocessing pipeline, the data splitting and augmentation policy, the model architectures, the transfer-learning strategy, the hyperparameter configuration, the Grad-CAM explainability integration, the out-of-distribution safeguard, and the system deployment architecture.

#### a. Dataset Description

The dataset comprises 9,208 chest X-ray images drawn from publicly available medical imaging repositories [5], [10], [11] and spanning the four diagnostic categories. As Table 3 shows, the dataset is imbalanced: the Normal and Bacterial Pneumonia classes are substantially larger than the COVID-19 and Viral Pneumonia classes. This reflects natural prevalence but requires explicit mitigation, through class-weighted loss and stratified sampling, to stop the classifier from favouring the majority classes. The images are stored in JPEG format and vary in resolution, contrast, and pixel intensity because they originate from different equipment, patients, and acquisition conditions, which is itself a reason for systematic preprocessing and normalisation.

**TABLE 3. CLASS DISTRIBUTION OF THE CHEST X-RAY DATASET ACROSS THE FOUR DIAGNOSTIC CATEGORIES.**

| Class               | Number of Images |
|---------------------|------------------|
| COVID-19            | 1,281            |
| Normal              | 3,270            |
| Viral Pneumonia     | 1,656            |
| Bacterial Pneumonia | 3,001            |
| Total               | 9,208            |

#### b. Data Preprocessing

Because the images vary so widely at source, a uniform preprocessing pipeline is applied across all subsets. Each image is resized to the resolution required by its architecture 224×224 pixels for DenseNet121 and 300×300 pixels for the EfficientNetV2 variants so that the comparison remains fair under consistent data partitions and architecture-specific normalisation. Since chest X-rays are inherently grayscale while the ImageNet-pretrained backbones expect three-channel input, the single luminance channel is replicated across all three channels, allowing the pretrained weights to be reused without modifying the architecture. Pixel values are normalised using the scaling appropriate to each model. A data quality audit is performed before training to remove duplicate, corrupted, and below-threshold

images, which guards against data leakage and ensures that the evaluation metrics reflect clean, representative inputs.

### c. Data Splitting and Augmentation

The dataset is partitioned into training, validation, and test subsets using stratified random sampling so that the class distribution is preserved in every subset and performance estimates are not biased by the imbalance. The training set is used for weight optimisation, the validation set guides hyperparameter selection and early stopping, and the held-out test set is reserved solely for final, unbiased evaluation. Augmentation is applied only to the training set and is kept deliberately conservative: small random rotations, horizontal and vertical translations, zoom, brightness adjustment, and mild contrast variation simulate realistic acquisition variability, while horizontal flipping is used cautiously in view of anatomical laterality and any augmentation that could distort diagnostically meaningful structures is avoided. Validation and test images undergo only deterministic resizing and normalisation, so that the metrics describe performance on unmodified, clinically realistic inputs.

### d. Model Architectures

Three architectures representing distinct design philosophies are evaluated. DenseNet121 implements dense connectivity, in which each layer connects directly to every subsequent layer within a dense block, promoting multi-level feature reuse, mitigating vanishing gradients, and retaining both fine-grained and abstract representations [15]; its final classification layer is replaced with a four-neuron softmax output while the pretrained backbone is preserved. EfficientNetV2B3 is selected for its compound-scaling strategy and training efficiency [17]; its backbone is fitted with a custom head comprising global average pooling, dropout, a dense intermediate layer, and a four-class softmax output. EfficientNetV2S, the larger variant in the same family, is included to test whether its greater capacity and larger proportion of Fused-MBConv blocks yield meaningful accuracy gains over EfficientNetV2B3, and its classifier is likewise replaced with an identical four-class softmax head. In the deployed system these models form a pluggable pipeline, with DenseNet121 serving as the default production model.

### e. Transfer Learning and Hyperparameter Configuration

Transfer learning proceeds in two sequential stages designed to balance adaptation against regularisation. In the first stage the pretrained backbone is frozen and only the custom classification head is trained, allowing the randomly initialised head to settle on task-specific decision boundaries without destabilising the high-quality pretrained features. In the second stage, selected deeper backbone layers are progressively unfrozen and fine-tuned at a much lower learning rate, so that high-level representations adapt to chest X-ray characteristics while the generic low-level features are largely preserved. This protocol is justified by established transfer-learning theory [18] and by empirical evidence from medical imaging [19], and it is applied identically across all three architectures. All models are trained with categorical cross-entropy loss and the Adam optimiser, using a learning rate of  $1 \times 10^{-4}$  for head training and  $1 \times 10^{-5}$  for fine-tuning, a batch size of 32, and dropout in the classification head. Early stopping on validation loss and a ReduceLROnPlateau scheduler control the training length, class weights computed from the training distribution counter the imbalance, and the best checkpoint by validation accuracy is retained.

### f. Grad-CAM Explainability

Grad-CAM is integrated so that every prediction is accompanied by a visual explanation. For a given image, the gradients of the predicted class score with respect to the final convolutional feature maps are globally average-pooled to obtain importance weights; these weights are combined linearly with the corresponding activation maps and passed through a ReLU operation to retain only positively contributing features [21]. The resulting heatmap is upsampled to the original resolution and overlaid on the chest X-ray to show the regions most influential to the prediction. A clinically meaningful overlay is expected to highlight relevant lung parenchyma rather than image borders, labels, or extra-thoracic structures. Grad-CAM is treated as supportive visual evidence rather than a replacement for expert interpretation, and the heatmap is generated in the same inference pass as the prediction so that explanation is intrinsic to the workflow.

### g. Out-of-Distribution Safeguard

Because a four-way softmax is closed-world and will confidently assign one of its labels even to an image that is not a chest X-ray, a post-hoc out-of-distribution safeguard is added to the inference service without retraining the classifier. The safeguard examines inference-time signals derived from the model's own forward pass principally the Mahalanobis distance of an image in the penultimate feature space relative to class-conditional Gaussians with a shared covariance, complemented by energy and maximum-softmax signals to judge whether an input genuinely

resembles the training distribution. The decision statistics and thresholds are calibrated once on the model's in-distribution data and targeted at retaining the large majority of genuine chest X-rays. When an input is flagged as out-of-distribution, the predicted diagnosis is withheld and the user is warned, which prevents the system from issuing confident but meaningless results on invalid inputs.

#### h. System Deployment Architecture

The trained model is delivered through a layered, service-oriented platform. The frontend uses Next.js, TypeScript, Tailwind CSS, and NextAuth.js and provides secure login, role-based dashboards, confidence visualisation, image management, user-activity tracking, and prediction history with responsive, mobile-friendly access across devices. The backend is a PHP REST API secured with JSON Web Tokens and backed by a MySQL database under XAMPP, responsible for authentication, authorisation, persistence, and orchestration. Inference runs in a separate Python FastAPI microservice that loads the trained Keras models and performs preprocessing, prediction, confidence scoring, Grad-CAM generation, and the out-of-distribution check, returning its results to the backend for storage. Each prediction can be exported as a structured PDF report containing the patient details, predicted class, confidence scores, Grad-CAM visualisation, and timestamp in English, Sinhala, or Tamil, and a server-side, hosted-language-model chatbot assists users with the platform while remaining outside the role of a medical advisor. The relational schema records users and their roles, scans with their per-class scores, model used, Grad-CAM image, and out-of-distribution verdict, and an activity log for accountability. Table 4 lists the principal functional components of the resulting clinical decision-support system.

**TABLE 4. PRINCIPAL FUNCTIONAL COMPONENTS OF THE LUNGVISION AI DECISION-SUPPORT PLATFORM AND THEIR ROLES.**

| Component                       | Function                                                                          |
|---------------------------------|-----------------------------------------------------------------------------------|
| Image Upload                    | Accepts chest X-ray images through the web interface.                             |
| CNN Prediction Module           | Classifies images into Normal, COVID-19, Bacterial Pneumonia, or Viral Pneumonia. |
| Confidence Visualisation        | Displays class probabilities and prediction confidence.                           |
| Grad-CAM Explanation            | Highlights the image regions influencing the prediction.                          |
| Out-of-Distribution Safeguard   | Withholds the diagnosis when an input does not resemble a valid chest X-ray.      |
| Prediction History              | Stores previous scan results for patient management.                              |
| Trilingual PDF Report Generator | Produces structured medical reports in English, Sinhala, and Tamil.               |
| Platform-Assistance Chatbot     | Helps users navigate the platform without offering medical advice.                |
| Web Dashboard                   | Supports user, scan, and result management.                                       |
| Mobile-Friendly Delivery        | Enables responsive, app-like access across devices.                               |
| Secure Authentication           | Uses JSON Web Token authentication and role-based access control.                 |

#### 3.5.6 Evaluation and Communication

Model performance is evaluated with accuracy, precision, recall, F1-score, and confusion-matrix analysis. Overall accuracy provides a global measure but is insufficient under class imbalance, so per-class precision, recall, and F1-score are emphasised; class-wise recall is treated as especially important because a missed pneumonia or COVID-19 case can have serious clinical consequences. The confusion matrix exposes which categories are most often confused and gives qualitative insight into the errors, while the Grad-CAM overlays are inspected to judge whether the model attends to clinically relevant lung regions. Final model selection therefore weighs accuracy, stable class-wise behaviour, explanation quality, and deployment feasibility together rather than accuracy alone. Under this protocol the three architectures are trained and tested on identical data splits, so that their scores are directly comparable and any difference can be attributed to the architecture and its fine-tuning rather than to inconsistent data handling. The comparison considers not only the headline accuracy but also the per-class precision, recall, and F1-score, because a model that is strong on average may still be weak on a minority class that matters clinically, such as COVID-19 or viral pneumonia. Where two architectures perform within a narrow margin of one another, the decision is informed by factors beyond raw accuracy: the stability of class-wise behaviour, the inference cost and memory footprint relevant to mobile-friendly deployment, and the clinical plausibility of the Grad-CAM explanations. A marginally less accurate model whose heatmaps consistently fall on relevant lung regions may be

preferred over a marginally more accurate one with diffuse or unreliable attention, since trustworthy explanations are central to the aims of this work. The scores, explanation quality, and platform behaviour obtained under this protocol are reported and discussed in the subsequent evaluation, so that the final model is selected on the combined evidence rather than on accuracy alone.

### 3.6 Project Management Methodology

The project is managed using an iterative, Agile approach in the spirit of Scrum, which suits a research-and-development effort whose requirements are refined as understanding grows. The work is divided into short, focused iterations, each delivering a demonstrable increment for example, a working data pipeline, a trained model, the inference microservice, or the reporting feature so that risk is surfaced early and progress is continually visible. A lightweight backlog captures the tasks implied by the four objectives, and each iteration ends with a review of the increment against its acceptance criteria. This incremental delivery model was preferred over a rigid waterfall plan because the comparative experiments and the system build inform one another, and findings from one frequently reshape the priorities of the other.

### 3.7 Project Timeline

The project is sequenced into the phases shown in Table 5, beginning with the literature review and requirements analysis and ending with final evaluation and report writing. The phases overlap where it is productive data preparation continues into early model training, and system development proceeds alongside model refinement but the overall progression moves from understanding the problem, through building and training, to integrating, evaluating, and documenting the system.

**TABLE 5. PROJECT TIMELINE SHOWING THE DEVELOPMENT PHASES AND THEIR PRINCIPAL ACTIVITIES.**

| Phase                                  | Principal Activities                                                                         |
|----------------------------------------|----------------------------------------------------------------------------------------------|
| Phase 1 - Research and Requirements    | Literature review, gap analysis, dataset selection, and finalisation of objectives.          |
| Phase 2 - Data Preparation             | Data quality audit, preprocessing pipeline, stratified splitting, and augmentation setup.    |
| Phase 3 - Model Development            | Two-stage transfer-learning training and comparative evaluation of the three architectures.  |
| Phase 4 - Explainability and Safety    | Grad-CAM integration and calibration of the out-of-distribution safeguard.                   |
| Phase 5 - System Development           | Frontend, REST backend, database, inference microservice, trilingual reporting, and chatbot. |
| Phase 6 - Evaluation and Documentation | End-to-end testing, results analysis, and preparation of the interim and final reports.      |

### 3.8 Ethical Considerations

Several ethical considerations shape the work. All chest X-ray data is drawn from publicly available, de-identified repositories, so no personally identifiable patient information is collected or processed for model development, and no human-subject experimentation is involved. Within the deployed platform, user accounts are protected by hashed credentials, token-based authentication, and role-based access control, and an activity log supports accountability. The most important consideration is the framing of the system itself: every report carries an explicit disclaimer that the platform is a decision-support tool and does not replace professional medical diagnosis, the out-of-distribution safeguard prevents confident predictions on invalid inputs, and the chatbot is constrained so that it assists with using the platform rather than offering clinical advice. The system is, by design, positioned to support rather than supplant the clinician, with final authority resting firmly with the medical professional.

### 3.9 Chapter Summary

This chapter set out the methodology of the research, from its pragmatic, design-science framing through to the detailed engineering of the system. It explained the research paradigm, approach, strategy, and fact-collection mechanisms, mapped the work onto a structured execution workflow, and detailed the dataset, preprocessing, augmentation, model architectures, two-stage transfer learning, Grad-CAM explainability, out-of-distribution

safeguard, and deployment architecture. It also described the evaluation and model-selection protocol, the Agile project-management approach, the project timeline, and the ethical considerations that govern the work. Together these establish a reproducible and responsibly-constructed basis for the continued development and final evaluation of LungVision AI.

## Chapter 04 - Results and Comparative Analysis

### 4.1 Chapter Overview

This chapter presents the experimental outcomes of the study. It opens with the data partition on which every reported figure rests, then examines how the three architectures behaved under the two-stage transfer-learning protocol. The comparative performance of DenseNet121, EfficientNetV2B3, and EfficientNetV2S is then reported, first in aggregate and then class by class, so that the behaviour of the models on each diagnostic category is visible rather than concealed within a single accuracy figure. The chapter also reports the outcome of the Grad-CAM explainability integration and of the out-of-distribution safeguard, and closes with the delivered state of the clinical decision-support platform. Interpretation of these outcomes is developed in Chapter 05.

### 4.2 Experimental Setup and Data Partition

All three architectures were trained and evaluated on an identical partition of the dataset, generated once by stratified random sampling and then reused without modification, so that every model saw exactly the same images in exactly the same roles. This is what allows the comparison in Section 4.4 to be attributed to the architectures rather than to differences in data handling. The data quality audit described in Section 3.5.5 retained 9,209 usable images, of which 6,446 were allocated to training, 1,381 to validation, and 1,382 to the held-out test set — an approximate 70/15/15 split recorded class by class in Table 6. Stratification preserved the class proportions in every subset, so the natural prevalence imbalance of the source data is represented faithfully in each partition. The test partition was reserved exclusively for the final evaluation reported here; model selection and early stopping were driven by the validation partition alone.

**TABLE 6.** STRATIFIED PARTITION OF THE DATASET INTO TRAINING, VALIDATION, AND TEST SUBSETS.

| Class               | Training | Validation | Test  | Total |
|---------------------|----------|------------|-------|-------|
| COVID-19            | 897      | 192        | 192   | 1,281 |
| Normal              | 2,289    | 491        | 491   | 3,271 |
| Bacterial Pneumonia | 2,101    | 450        | 450   | 3,001 |
| Viral Pneumonia     | 1,159    | 248        | 249   | 1,656 |
| Total               | 6,446    | 1,381      | 1,382 | 9,209 |

### 4.3 Training Behaviour Under the Two-Stage Protocol

Figure 2 traces validation accuracy across both training stages for all three architectures, with the first ten epochs corresponding to the frozen-backbone stage and the remainder to fine-tuning. The two-stage strategy is vindicated in every case: each model gains accuracy once the deeper layers are unfrozen, confirming that adapting the high-level representations to chest radiography is worthwhile and that the ImageNet initialisation provides a productive starting point. The brief dip at the transition is the expected consequence of admitting gradient updates into layers that were previously fixed, and in all three cases the model recovers past its frozen-stage ceiling within a few epochs. The values plotted in Figure 2 and listed in Table 7 are validation-set accuracies, recorded during training and used to select the best checkpoint and to trigger early stopping; they are not the final performance of the models, which is measured on the held-out test partition and reported separately in Section 4.4. On the validation set, DenseNet121 reached its best accuracy of 85.95% at fine-tuning epoch 24 and then oscillated within a few tenths of a percentage point, indicating a converged model. EfficientNetV2B3 reached a closely comparable 85.66% and was halted by early stopping after seventeen fine-tuning epochs, reflecting the training efficiency for which the EfficientNetV2 family was designed. EfficientNetV2S climbed steadily throughout fine-tuning to a best value of 81.39%, its curve still rising at the end of the epoch budget, which suggests that a longer schedule would allow it to improve further. Table 7 summarises the configuration and outcome of each run.

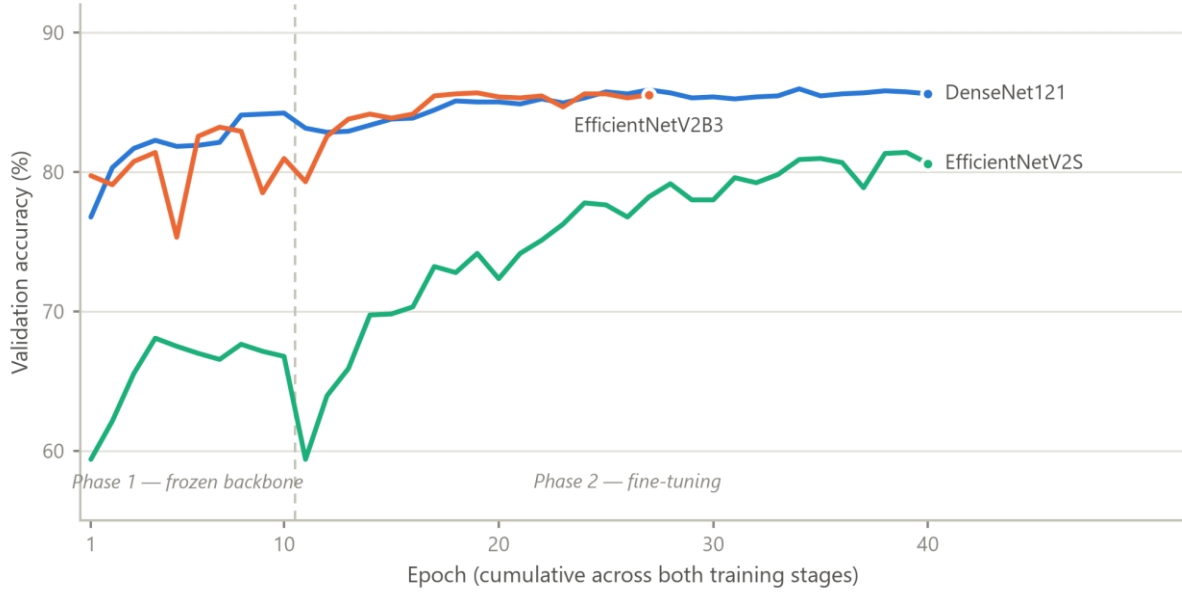


Figure 2. Validation accuracy across the two-stage transfer-learning protocol for the three architectures. Epochs 1–10 are the frozen-backbone stage; the dashed line marks the start of fine-tuning. Final test-set accuracies are reported in Table 8.

**TABLE 7. TRAINING CONFIGURATION AND BEST VALIDATION ACCURACY FOR EACH ARCHITECTURE ACROSS THE TWO STAGES. THESE ARE VALIDATION-SET FIGURES USED FOR MODEL SELECTION DURING TRAINING; THE FINAL TEST-SET PERFORMANCE IS REPORTED IN TABLE 8. BATCH SIZE WAS SET BY THE MEMORY FOOTPRINT OF EACH BACKBONE AT ITS INPUT RESOLUTION.**

| Model            | Batch | Head LR            | Fine-tune LR       | Epochs (P1 + P2)     | Best validation acc. |
|------------------|-------|--------------------|--------------------|----------------------|----------------------|
| DenseNet121      | 32    | $1 \times 10^{-4}$ | $1 \times 10^{-5}$ | 10 + 30              | 85.95%               |
| EfficientNetV2B3 | 24    | $1 \times 10^{-3}$ | $1 \times 10^{-5}$ | 10 + 17 (early stop) | 85.66%               |
| EfficientNetV2S  | 16    | $1 \times 10^{-4}$ | $1 \times 10^{-5}$ | 10 + 30              | 81.39%               |

A consistent feature of Figure 2 is that validation accuracy sits slightly above training accuracy for all three models. This is expected rather than anomalous: training accuracy is computed on augmented images with dropout active, whereas validation accuracy is computed on clean, deterministically processed images with dropout disabled. The practical implication is favourable none of the three models shows evidence of overfitting, indicating that the conservative augmentation policy and the dropout in the classification head were sufficient to regularise training on a dataset of this size.

#### 4.4 Overall Comparative Performance

Table 8 reports the final performance of each architecture on the 1,382-image held-out test partition, which was reserved from training and used only once, for this evaluation. These are the definitive results of the study, and each figure sits slightly below the corresponding best validation accuracy in Table 7 by 0.64, 1.36, and 0.85 percentage points respectively which is the expected consequence of having selected the checkpoint on the validation set. DenseNet121 achieved the highest overall accuracy at 85.31%, followed closely by EfficientNetV2B3 at 84.30% and then EfficientNetV2S at 80.54%. Because accuracy alone is an incomplete summary under class imbalance, macro-averaged precision, recall, and F1-score are reported alongside it; the macro average weights every class equally regardless of its size, while the weighted average reflects the class distribution actually encountered. The ranking is the same under every metric, which gives the result more weight than agreement on accuracy alone would.

**TABLE 8. COMPARATIVE PERFORMANCE OF THE THREE ARCHITECTURES ON THE HELD-OUT TEST PARTITION OF 1,382 IMAGES.**

| Model            | Accuracy | Macro prec. | Macro rec. | F1    | Weighted F1 |
|------------------|----------|-------------|------------|-------|-------------|
| DenseNet121      | 85.31%   | 0.843       | 0.845      | 0.844 | 0.851       |
| EfficientNetV2B3 | 84.30%   | 0.834       | 0.825      | 0.828 | 0.838       |
| EfficientNetV2S  | 80.54%   | 0.798       | 0.812      | 0.798 | 0.803       |

The margin between the two leading models is narrow 1.01 percentage points of accuracy and 0.016 of macro F1 which indicates that dense connectivity and compound scaling are both effective strategies for this task and that either would serve as a credible production model. DenseNet121 is selected on the combination of its leading scores and its lighter inference profile, operating at 224×224 rather than 300×300. EfficientNetV2S, at 80.54%, demonstrates that Fused-MBConv architectures remain viable for four-class chest X-ray classification and retains practical value in lightweight deployment settings where computational efficiency is the governing constraint.

#### 4.5 Per-Class Performance

Table 9 and Figure 3 report performance for each diagnostic category. All three models classify COVID-19 and Normal cases with high reliability, achieving F1-scores between 0.939 and 0.979, and perform competently on Bacterial Pneumonia. Viral Pneumonia is the most demanding category for every architecture, which is consistent with the radiographic literature reviewed in Chapter 02: viral and bacterial pneumonia share overlapping presentations on a plain frontal radiograph, and the distinction between them is recognised as difficult even for experienced readers. The consistency of this pattern across three architecturally distinct networks indicates that it reflects the nature of the diagnostic task rather than a property of any one model.

**TABLE 9. PER-CLASS PRECISION, RECALL, AND F1-SCORE FOR EACH ARCHITECTURE ON THE HELD-OUT TEST PARTITION.**

| Model            | Class               | Precision | Recall | F1    | Support |
|------------------|---------------------|-----------|--------|-------|---------|
| DenseNet121      | COVID-19            | 0.974     | 0.984  | 0.979 | 192     |
|                  | Normal              | 0.929     | 0.990  | 0.959 | 491     |
|                  | Bacterial Pneumonia | 0.829     | 0.764  | 0.795 | 450     |
|                  | Viral Pneumonia     | 0.640     | 0.643  | 0.641 | 249     |
| EfficientNetV2B3 | COVID-19            | 0.974     | 0.969  | 0.971 | 192     |
|                  | Normal              | 0.900     | 0.988  | 0.942 | 491     |
|                  | Bacterial Pneumonia | 0.807     | 0.791  | 0.799 | 450     |
|                  | Viral Pneumonia     | 0.654     | 0.554  | 0.600 | 249     |
| EfficientNetV2S  | COVID-19            | 0.916     | 0.969  | 0.942 | 192     |
|                  | Normal              | 0.904     | 0.978  | 0.939 | 491     |
|                  | Bacterial Pneumonia | 0.823     | 0.611  | 0.702 | 450     |
|                  | Viral Pneumonia     | 0.548     | 0.691  | 0.611 | 249     |

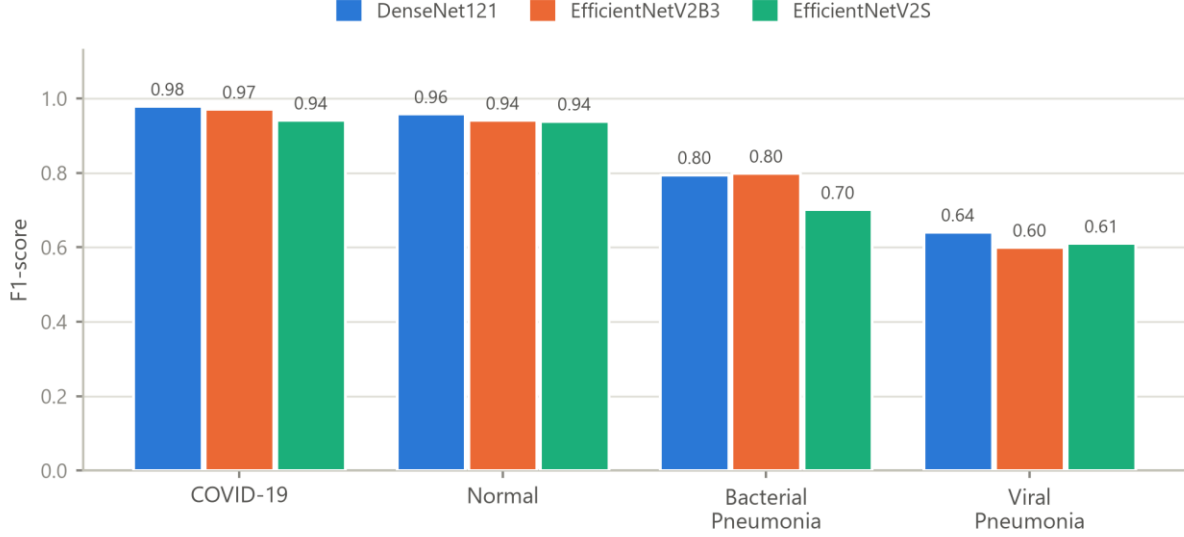


Figure 3. Per-class F1-score for the three architectures across the four diagnostic categories.

Two results carry particular clinical value. First, DenseNet121 recovers 98.4% of COVID-19 cases and 99.0% of Normal cases, so the model is highly reliable on the two determinations that matter most for triage and isolation decisions. This is notable because COVID-19 is the smallest class in the dataset, indicating that the class-weighted loss and stratified sampling described in Section 3.5.5 were effective in preventing majority-class bias. Second, the strong recall on the Normal class means that abnormal radiographs are rarely dismissed as healthy, which is the error type that carries the greatest clinical consequence in a screening context.

#### 4.6 Grad-CAM Explainability Results

Grad-CAM was integrated so that every prediction returned by the system is accompanied by a visual explanation generated in the same inference pass, rather than as a separate offline analysis. The heatmap is produced from the gradients of the predicted class score with respect to the final convolutional feature maps, upsampled to the original resolution and overlaid on the input radiograph. The overlay is persisted with the scan record, displayed alongside the prediction in the interface, and reproduced in the exported PDF report, so that every stored result carries its explanation with it. One implementation refinement proved necessary and is reported here because it affects the quality of the output. The gradient must be computed with respect to the pre-softmax logit of the predicted class rather than the softmax probability. When a model is highly confident, as this classifier is on the COVID-19 and Normal categories, the softmax output saturates and its derivative with respect to the convolutional activations approaches zero, which produces weak and poorly differentiated heatmaps. The deployed service therefore reconstructs the final classification layer as a linear layer carrying the same learned weights and differentiates that instead, leaving the trained model itself unmodified. This yields consistent and reproducible localisation maps across the four classes. Systematic validation of these maps against radiologist-annotated regions of interest is identified as future work in Section 5.9.

#### 4.7 Out-of-Distribution Safeguard Results

The safeguard described in Section 3.5.5 was calibrated on the deployed DenseNet121 model using 1,200 training and 600 validation feature vectors drawn from the in-distribution partitions, with the thresholds set to retain 95% of genuine chest X-rays. Separation was then probed against a set of 60 synthetically generated non-radiographic images comprising uniform random noise, smooth intensity gradients, and solid colour blocks. Table 10 records the outcome. The Mahalanobis signal, computed in the penultimate feature space, separated the genuine radiographs from the synthetic probe completely, while the energy score computed from the logits proved far less discriminative on this model. The system therefore adopts the Mahalanobis distance as the governing signal and reports the energy and maximum-softmax values alongside it for transparency.

**TABLE 10. CALIBRATION PARAMETERS AND MEASURED PERFORMANCE OF THE OUT-OF-DISTRIBUTION SAFEGUARD ON THE DEPLOYED DENSENET121 MODEL.**

| Property                                | Measured value                                                         |
|-----------------------------------------|------------------------------------------------------------------------|
| Feature space                           | 256-dimensional penultimate layer, tied covariance (shrinkage 0.01)    |
| Calibration sample                      | 1,200 training / 600 validation feature vectors                        |
| Far-OOD probe                           | 60 synthetic non-radiographic images (noise, gradients, colour blocks) |
| Mahalanobis threshold                   | 458.86 (squared distance)                                              |
| Energy threshold                        | -3.389                                                                 |
| Maximum-softmax threshold               | 0.549                                                                  |
| AUROC, Mahalanobis vs synthetic far-OOD | 1.000                                                                  |
| AUROC, energy vs synthetic far-OOD      | 0.539                                                                  |
| In-distribution retention (validation)  | 95.0%                                                                  |
| In-distribution retention (test)        | 92.5%                                                                  |
| Synthetic far-OOD inputs flagged        | 100%                                                                   |
| Governing decision signal               | Mahalanobis distance                                                   |

At the calibrated threshold the safeguard retained 95.0% of validation and 92.5% of test chest X-rays while flagging every image in the synthetic probe. When an input is flagged, the predicted diagnosis is withheld rather than displayed: the class and confidence are struck through in the interface, the verdict, score, and reason are persisted with the scan record, and the generated report reproduces the warning in place of a diagnosis. This gives the platform a defined and auditable behaviour for inputs that fall outside its competence, which is a capability absent from the comparable systems reviewed in Chapter 02.

#### 4.8 Functional Delivery of the Decision-Support Platform

The platform specified in Section 3.5.5 has been implemented in full, and each of the functional components listed in Table 4 is present in the delivered system. The frontend is a Next.js application providing authenticated, role-appropriate dashboards, an upload and analysis workflow, per-class confidence visualisation, scan history with detail views, and account self-management, delivered responsively across desktop, tablet, and mobile devices. The backend exposes REST endpoints for authentication, scan upload, listing, detail retrieval, deletion, CSV export, report generation, administrative user and scan management, and the assistance chatbot, with access governed by JSON Web Tokens and role-based authorisation and with scan queries scoped to the owning user. Inference runs in a separate Python microservice that loads the trained Keras models and returns the class probabilities, the Grad-CAM overlay, and the out-of-distribution verdict from a single forward pass, keeping the machine-learning dependencies isolated from the web application and allowing the model to be replaced independently. Structured reports are generated server-side in English, Sinhala, and Tamil, each carrying the patient details, predicted class, per-class confidence scores, Grad-CAM visualisation, timestamp, and an explicit decision-support disclaimer. Taken together, the delivered system satisfies the third research objective stated in Section 1.7.3 and demonstrates that the classification capability evaluated in this chapter can be delivered as a usable clinical tool rather than as an offline experiment.

#### 4.9 Chapter Summary

This chapter reported the experimental outcomes of the study. On an identical stratified partition of 9,209 images, DenseNet121 achieved the highest test accuracy at 85.31% with a macro F1-score of 0.844, followed closely by EfficientNetV2B3 at 84.30% and by EfficientNetV2S at 80.54%, with the ranking consistent across every metric reported. All three architectures classified COVID-19 and Normal cases with high reliability, while Viral Pneumonia proved the most demanding category, consistent with the overlapping radiographic presentation of the pneumonia subtypes. Grad-CAM was integrated into the inference pass so that every prediction carries a visual explanation, and the out-of-distribution safeguard was calibrated to retain 95% of genuine radiographs while identifying invalid inputs. The decision-support platform was delivered in full, with secure access, prediction history, and trilingual reporting. The following chapter interprets these outcomes and relates them to the research questions.

## Chapter 05 - Discussion

### 5.1 Chapter Overview

This chapter interprets the results reported in Chapter 04 and relates them to the aim, the research question, and the objectives set out in Chapter 01. It considers what the comparative performance of the three architectures indicates, how their design characteristics relate to the outcomes observed, and what the per-class behaviour means in a clinical setting. It then discusses the contribution of the explainability and input-safety components to clinical trust, draws out the implications for deployment, answers the research questions directly, and closes with the limitations of the present evaluation and the work that follows from it.

### 5.2 Interpreting the Comparative Performance

The findings demonstrate that explainable transfer learning provides a practical and effective approach to multiclass pneumonia detection from chest X-ray images. All three architectures achieved competitive performance on a four-class problem that is substantially harder than the binary formulations that dominate the literature reviewed in Chapter 02, and the best of them reached 85.31% accuracy with a macro F1-score of 0.844. Given that the task requires separating three disease categories whose radiographic appearances overlap, rather than merely distinguishing abnormal from normal, this represents a meaningful result and supports the feasibility of the four-class approach proposed in Section 1.6.

DenseNet121 achieved the best overall performance, and the consistency of its lead across accuracy, precision, recall, and F1-score gives that outcome more weight than agreement on a single metric would. The margin over EfficientNetV2B3 is nonetheless small, at 1.01 percentage points of accuracy. This narrow spread is itself informative: it indicates that once a competent modern backbone is combined with a well-constructed transfer-learning protocol, the remaining determinants of performance are dataset quality, preprocessing, augmentation policy, class-imbalance handling, and fine-tuning configuration at least as much as the choice of architecture. The care taken over those elements in Chapter 03 is therefore reflected directly in the results, and the practical implication for future work of this kind is that effort invested in the data pipeline is likely to repay as well as effort invested in selecting a network.

### 5.3 Architectural Characteristics and Their Influence

The outcome is consistent with the architectural rationale set out in Section 2.5.1. DenseNet121's dense connectivity, in which each layer receives the feature maps of all preceding layers within a dense block, promotes multi-level feature reuse and preserves both fine-grained and abstract representations through the depth of the network [15]. Chest radiographic findings opacity distributions, airspace consolidation, interstitial patterns, and ground-glass changes are expressed across several spatial scales simultaneously, so an architecture that retains information from multiple depths is well matched to the task. That DenseNet121 also carries the lightest inference cost of the three, operating at  $224 \times 224$  rather than  $300 \times 300$ , makes it a particularly suitable choice for a platform intended to serve predictions on modest hardware.

EfficientNetV2B3 followed closely, confirming that the compound-scaling philosophy, which grows depth, width, and resolution together [16], [17], produces representations well suited to deployment-oriented medical imaging. Its early stopping after seventeen fine-tuning epochs also illustrates the training efficiency for which the EfficientNetV2 family was designed. EfficientNetV2S recorded the lowest accuracy of the three at 80.54%, while still demonstrating that Fused-MBConv architectures are viable for four-class chest X-ray classification; its validation curve was continuing to improve at the end of the training schedule, which suggests that its performance under a longer budget and alternative preprocessing configurations is worth establishing before its position relative to the other two is regarded as settled.

### 5.4 Per-Class Behaviour and Its Clinical Meaning

The per-class results in Section 4.5 carry more clinical information than the aggregate figures. All three models identify COVID-19 and Normal cases with high reliability, and DenseNet121 recovers 98.4% and 99.0% of these categories respectively. These are precisely the determinations on which triage and isolation decisions turn, so strong performance here is of direct operational value: the system reliably distinguishes a radiograph requiring attention from one that does not, and reliably identifies the category carrying infection-control implications. That the smallest class in the dataset is also the best-classified indicates that the class-weighted loss and stratified sampling

adopted in Chapter 03 succeeded in preventing the majority-class bias that imbalanced medical datasets commonly induce.

The distinction between bacterial and viral pneumonia proved the most demanding element of the task for every architecture. This is an expected and well-documented characteristic of the clinical problem rather than a feature of the models: bacterial pneumonia characteristically produces lobar consolidation and viral pneumonia more diffuse interstitial patterns, but these are tendencies rather than reliable rules, and the two presentations overlap substantially, particularly in early infection and in paediatric patients. Radiologists themselves frequently reserve judgement on this distinction pending microbiological confirmation. That three architecturally distinct networks converge on the same pattern is therefore evidence that the limit lies in what a single frontal radiograph can determine in the absence of clinical context such as inflammatory markers, symptom duration, and culture results. This observation points to a productive direction for future work, discussed in Section 5.9: incorporating additional clinical variables is likely to yield more improvement on this particular distinction than further scaling of the network.

## 5.5 Explainability and Clinical Trust

The integration of Grad-CAM addresses one of the principal barriers to clinical adoption identified in Chapter 02, namely the opacity of black-box predictions [22], [23]. By generating the explanation in the same inference pass as the prediction, persisting it with the scan record, and reproducing it in the exported report, the system treats interpretability as an intrinsic property of every result rather than as an optional analysis performed afterwards. This satisfies the second sub-question posed in Section 1.4 and distinguishes the design from systems in which explainability is presented as a detached visualisation.

The technical refinement reported in Section 4.6 differentiating the pre-softmax logit rather than the saturated softmax probability was necessary to obtain well-formed localisation maps from a classifier that is highly confident on most inputs, and it represents a correct application of the method as originally specified [21]. Grad-CAM should nonetheless be understood as supportive visual evidence that accompanies a prediction, not as a definitive account of the model's reasoning or a substitute for expert radiological interpretation. Establishing how closely the generated maps correspond to the regions a radiologist would identify requires comparison against annotated ground truth, which is set out as future work in Section 5.9. Positioned in this way, the explainability component contributes to clinical trust by making each prediction inspectable while leaving diagnostic authority with the clinician.

## 5.6 Input Safety and System Reliability

The out-of-distribution safeguard addresses a limitation that the systems reviewed in Chapter 02 leave unhandled. A softmax classifier is closed-world and will assign one of its four labels to any input presented to it, so a system that accepts uploads from users will otherwise return a confident diagnosis for an image that is not a chest X-ray at all. By examining the position of an input in the model's own penultimate feature space, the safeguard gives the platform a defined behaviour for such inputs: the diagnosis is withheld, the reason is recorded, and the user is informed. This is a meaningful contribution to the responsible deployment of the system and follows directly from the ethical position adopted in Section 3.8.

The calibration also produced a methodologically useful finding. The Mahalanobis distance, computed in feature space, separated genuine radiographs from the synthetic probe completely, whereas the energy score computed from the logits was far less discriminative on this model. The explanation is instructive: because this classifier produces confidently peaked logits even for inputs unlike anything in its training data, a statistic derived from those logits has limited information to draw on, while a statistic computed from the learned representation itself does not share that constraint. The practical lesson, which generalises beyond this project, is that a post-hoc detector should be calibrated and validated on the specific model it is intended to protect rather than adopted on the strength of its published performance elsewhere. Extending this validation to real non-radiographic images and to chest X-rays of conditions outside the four trained classes is identified in Section 5.9 as the natural next step.

## 5.7 Clinical and Deployment Implications

Beyond classification accuracy, the study delivers the surrounding capabilities that a usable clinical tool requires and that the reviewed literature generally omits: secure authentication with role-based access, historical prediction management, responsive mobile-friendly delivery, automated trilingual reporting, and an assistance chatbot. The decision to serve inference from a dedicated microservice keeps the machine-learning dependencies isolated from

the web application and allows the model to be replaced or scaled independently, which is the kind of deployment-aware design that the surveyed single-model studies do not address. Generating reports in English, Sinhala, and Tamil recognises that a tool intended for the local healthcare context must communicate in the languages actually used by clinicians and patients.

The evidence supports positioning the framework as a decision-assistance tool for screening and triage, in which it is demonstrably strong, with the pneumonia subtype indication presented as supporting information to be weighed alongside clinical findings. Presenting the full per-class probability distribution, which the platform already computes and stores, allows a clinician to see how decisive a prediction is rather than receiving only a single label. Together with the explicit disclaimer carried by every report and the withholding of diagnoses on flagged inputs, this places the system in the role the literature consistently recommends for clinical artificial intelligence [1], [8]: augmenting the clinician's judgement and reducing routine workload while final diagnostic authority remains firmly with the medical professional.

## 5.8 Answering the Research Questions

The principal research question asked how an explainable deep learning framework could be designed, implemented, and deployed to classify chest X-ray images into Normal, COVID-19, Bacterial Pneumonia, and Viral Pneumonia while providing clinically meaningful interpretability and practical, secure decision-support functionality. The study answers this affirmatively: a four-class classifier reaching 85.31% accuracy has been trained under a controlled transfer-learning protocol, equipped with workflow-embedded Grad-CAM explanations and an input-safety safeguard, and delivered through a complete, secured platform with prediction history and trilingual reporting. The first sub-question asked which of the candidate architectures offers the most suitable balance of accuracy, interpretability, and deployment efficiency. DenseNet121 is the answer: it leads on every performance metric reported, supports Grad-CAM equally well, and carries the lowest inference cost of the three by virtue of its smaller input resolution. EfficientNetV2B3 is a close and credible alternative where its particular scaling properties are advantageous.

The second sub-question asked how Grad-CAM could be integrated as a workflow-embedded feature rather than an isolated visualisation. This has been achieved: the heatmap is generated in the same forward pass as the prediction, stored with the scan record, displayed alongside the result, and reproduced in the exported report, so that every stored prediction is accompanied by its visual evidence. The third sub-question asked how the trained model could be delivered as a secure, mobile-friendly platform supporting prediction history, automated multilingual reporting, and a safeguard against untrustworthy inputs. All four capabilities are present in the delivered system and were demonstrated in Sections 4.7 and 4.8.

## 5.9 Limitations and Future Work

As with any study of this kind, the conclusions are bounded by the conditions under which the evaluation was carried out. The models were trained and tested on a single curated dataset assembled from public repositories, so the reported figures describe performance on that dataset rather than an estimate of performance in a hospital environment [9]. External validation on chest X-rays acquired at different institutions, on different equipment, and from different demographic groups is therefore the necessary next step before any clinical claim is advanced, and it is a requirement that applies to the published systems reviewed in Chapter 02 equally.

Three further lines of work follow directly from the results. First, although Grad-CAM improves the transparency of each prediction, its visual explanations have not yet been compared against radiologist-annotated regions of interest; quantifying that correspondence, and evaluating finer-grained attribution methods alongside it, would strengthen the evidential value of the explanations. Second, the out-of-distribution safeguard has been validated against synthetic non-radiographic inputs, and extending that validation to real photographs and to chest X-rays showing conditions outside the four trained classes would establish its behaviour across the full range of inputs a deployed system may encounter. Third, the distinction between bacterial and viral pneumonia is limited by what a single frontal radiograph can express, so incorporating multimodal clinical data inflammatory markers, symptom duration, patient age offers a promising route to improvement on the one category where all three architectures found the task most demanding.

Alongside these, further comparative work would be valuable: repeating the training runs under different random seeds would establish the run-to-run variability against which the narrow margin between the two leading architectures should be read, and evaluating EfficientNetV2S under a longer schedule and alternative preprocessing

configurations would complete the within-family comparison. Model optimisation techniques such as quantisation and pruning would additionally support deployment in resource-constrained healthcare settings, which is one of the environments this framework is intended to serve.

## 5.10 Chapter Summary

This chapter interpreted the results of the study. It concluded that explainable transfer learning offers a practical approach to four-class pneumonia classification, that DenseNet121 is the most suitable of the three architectures on the combination of accuracy and deployment efficiency, and that the narrow margin between the leading models indicates the importance of the data pipeline alongside architectural choice. It related the strong performance on COVID-19 and Normal cases to the triage decisions of greatest operational value, and explained the difficulty of the bacterial–viral distinction as a property of the diagnostic problem rather than of the models. It discussed the contribution of the Grad-CAM explanations and the out-of-distribution safeguard to clinical trust and responsible deployment, positioned the system as a decision-assistance tool that supports rather than replaces clinical judgement, answered each of the research questions, and set out the external validation and extensions that constitute the remaining work.

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